

EEE3094S 2022: Class Test 1

- The test is open book
- Answer on the test sheet. If you run out of space, you may attach additional pages, but please note that you have done so (e.g. write “see attached pages” by the question) and make sure they are marked with your student number.
- There are four questions totalling 20 marks. There are also two bonus marks available, but the total may not exceed 100%.
- You have one hour to complete the test.
- The last page is an information sheet.
- The information sheet includes a photo of my housemate’s cat, Krummel. You are invited to draw a picture of him if you finish/ give up early and are bored. Doing so will not officially get you extra marks, but it will probably make tutor marking your test happy, which might make a difference in your favour.

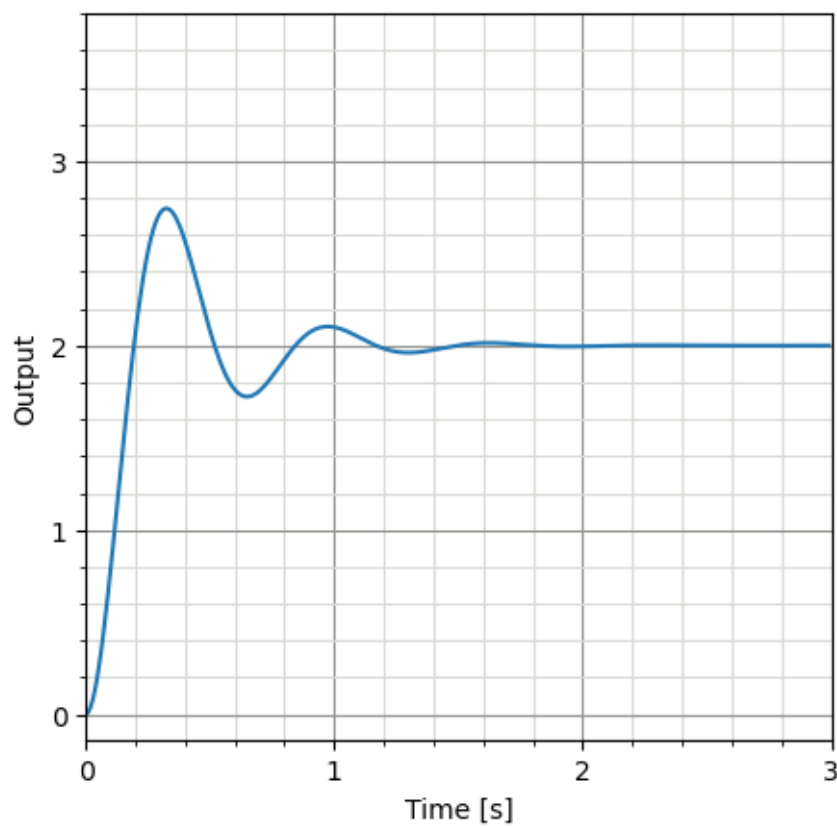
Question 1 [5]

Consider an unknown second-order system,

$$G(s) = \frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

The response of the system must satisfy the following specifications:

- Settling time to within 5% of the final value of less than 1 second.
 - Damping ratio greater than 0.5.
- a) Calculate the locations of the poles at the limits of these specifications. [2 marks]
- b) The unit step response of the system is shown below. Determine whether it satisfies these specifications. [3 marks]



Question 2 [5]

A system with the open-loop transfer function $G(s) = \frac{1}{s+2}$ is placed in closed loop to improve its ability to track a step input.

Two possible designs are proposed to control the closed-loop system:

Option 1: a proportional controller $C_1(s) = K$

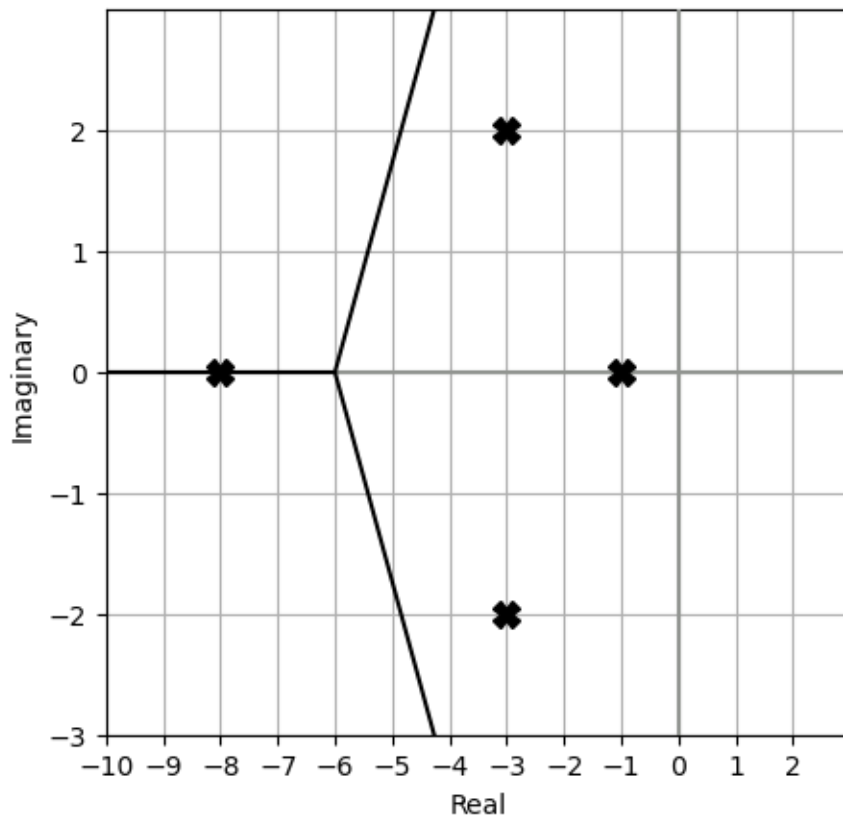
Option 2: an integral controller $C_2(s) = \frac{K}{s}$

The final design must be able to satisfy the following specifications:

- Greater than 95% error with respect to position (step) inputs.
 - Settling time not exceeding that of the open-loop system.
 - Damping ratio greater than 0.7
- a) Calculate the error constant of the uncontrolled closed-loop system with respect to a unit step input. [2 marks]
- b) Which of the controller options should be used? You may support your argument with calculations and/or rough sketches of the root locus. [3 marks]

Question 3 [5]

The plot below shows the open-loop poles of a system, and the asymptotes of its root locus (the black lines).



- Use this information to show that the system must have one zero located in the right-half plane at $s = 3$. [2 marks]
- True or false: the unit step response of the open-loop system will initially move away from the final value. Explain. [1 mark]
- Complete the root locus by showing where it exists on the real axis, and roughly sketching the trajectories of the poles. (You will need to redraw the plot.) [2 marks]

[illegible]

Question 4 [5]

A system with the open-loop transfer function $G(s) = \frac{1}{s+6}$ is required to track a velocity input (that is, a ramp) without error in closed loop.

Three controller designs are suggested:

Option 1: $C_1(s) = \frac{1}{s^2}$

Option 2: $C_2(s) = \frac{s+5}{s^2}$

Option 3: $C_3(s) = \frac{s+7}{s^2}$

- With reference to the type number of the system, explain why the controller must add at least two poles at the origin. [1 mark]
- Only one of the proposed controllers will result in an acceptable closed-loop system. Which one is it? Support your answer with root locus sketches for the three designs. [4 marks]
- BONUS: True or false: an underdamped second-order model is an acceptable approximation of the complete system with the working design. Explain. [2 marks]

Information

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\cos \theta = \zeta$$

$$t_p = \frac{\pi}{\omega_d}$$

$$y_p = A \left(1 + e^{-\zeta\pi / \sqrt{1-\zeta^2}} \right)$$

$$t_r = \frac{\pi - \theta}{\omega_d}$$

$$t_{k\%} = \frac{\ln \left(\frac{k}{100} \sqrt{1 - \zeta^2} \right)}{-\zeta\omega_n}$$

Laplace transforms of signals:

$y(t)$	$Y(s)$	Comments
$\sigma(t)$	$\frac{1}{s}$	unit step
$t\sigma(t)$	$\frac{1}{s^2}$	unit ramp
$e^{-at}\sigma(t)$	$\frac{1}{(s+a)}$	decaying exponential if $a > 0$
$t^n e^{-at}\sigma(t)$	$\frac{n!}{(s+a)^{(n+1)}}$	n^{th} order exponential
$\sin(\omega t)\sigma(t)$	$\frac{\omega}{s^2 + \omega^2}$	sinusoid
$\cos(\omega t)\sigma(t)$	$\frac{s}{s^2 + \omega^2}$	<u>cosinusoid</u>
$e^{-at}\sin(\omega t)\sigma(t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$	damped sinusoid
$e^{-at}\cos(\omega t)\sigma(t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$	damped <u>cosinusoid</u>



Fig. 1 Krummel wearing a bowtie